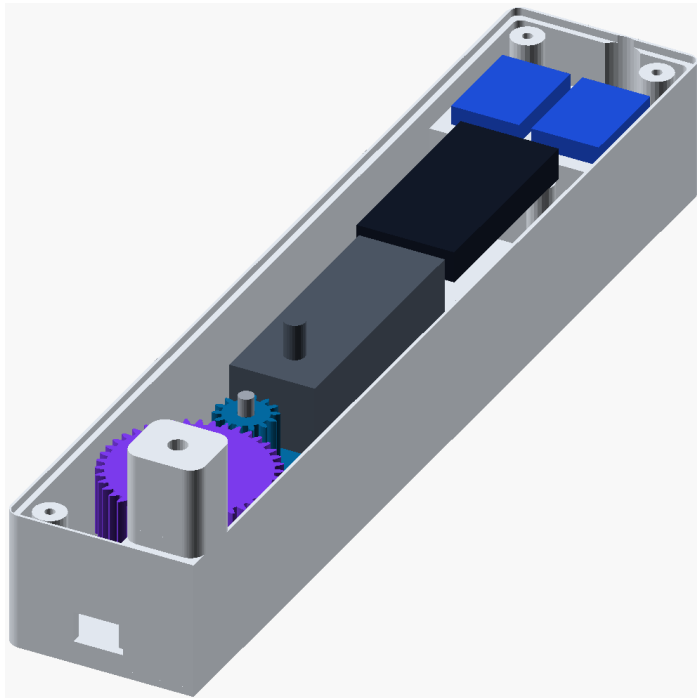
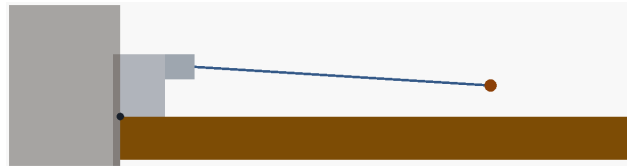


# doorbot

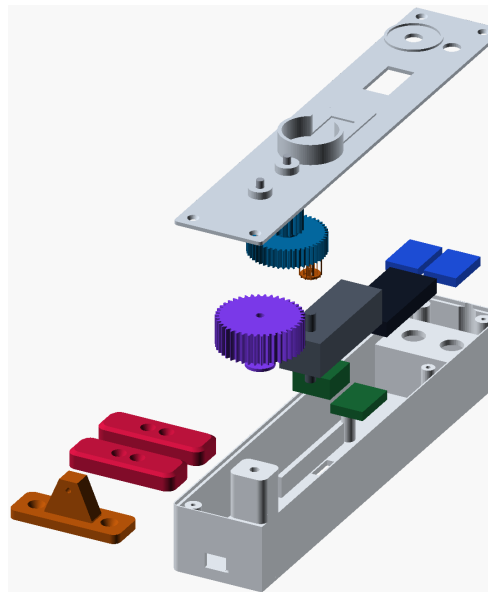
A door closer printed in Bambu Lab PLA Tough+, driven by one TT motor, run by a TTGO T-Display



assembled, cover on



plan view, shut: unit in the rebate, cable to the anchor



seven printed designs, eight pieces

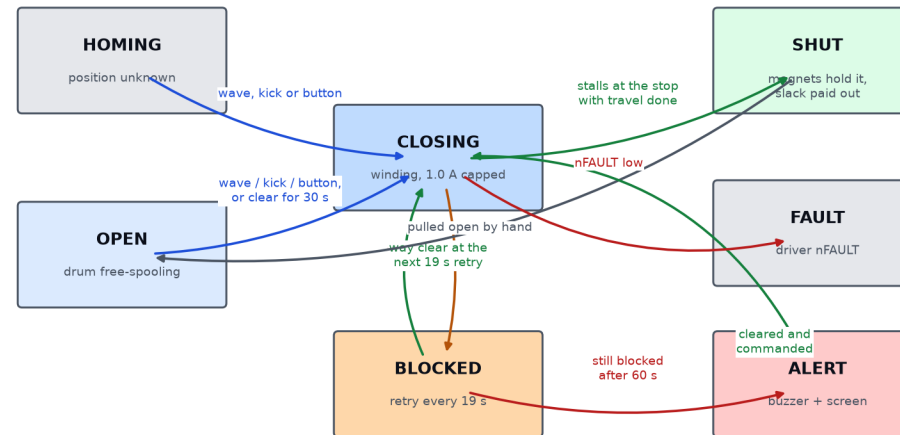
It screws to the hinge-side jamb, behind the door, and reels a Dyneema cable in to swing the door shut. Four magnets at the latch edge take the last millimetres and hold it closed. Wave a hand at it, kick the door twice, or let it close on its own when the doorway is clear.

|                      |                               |
|----------------------|-------------------------------|
| printed designs      | 7 (8 pieces)                  |
| printed material     | 256 g PLA Tough+              |
| purchased pieces     | 46 (\$124.49)                 |
| gear ratio           | 12:1, two printed spur stages |
| cable tension        | 151 N, capped in hardware     |
| moment arm           | 42-54 mm, never zero          |
| torque margin        | 2.98x worst case              |
| close time           | 21 s from wide open           |
| door it is sized for | 914 mm solid core, 40 kg      |
| magnet hold          | 70 N at a 2 mm gap            |

# What it does

Three ways to make it close, and one rule when something is in the way

## What it does, and when



Obstacle policy, exactly as specified: "Retry every 19 seconds for a minute, then alert". Three retries fit in the window (19 s, 38 s, 57 s); the fourth would fall outside it, so at 60 s it alerts instead. firmware/test/test\_controller.cpp asserts all of that on the host.

### 1 Wave a hand over the ring

Two consecutive readings inside 20-250 mm of the engraved target. One brush past does nothing.

### 3 Leave it alone

Doorway clear for 30 s and the door still open: it closes itself.

### 5 Just open the door

The drum free-spools. It costs about 1 N extra at the door edge, powered or not.

### 2 Kick the door twice

Two impacts 120-900 ms apart, felt through the jamb by the accelerometer.

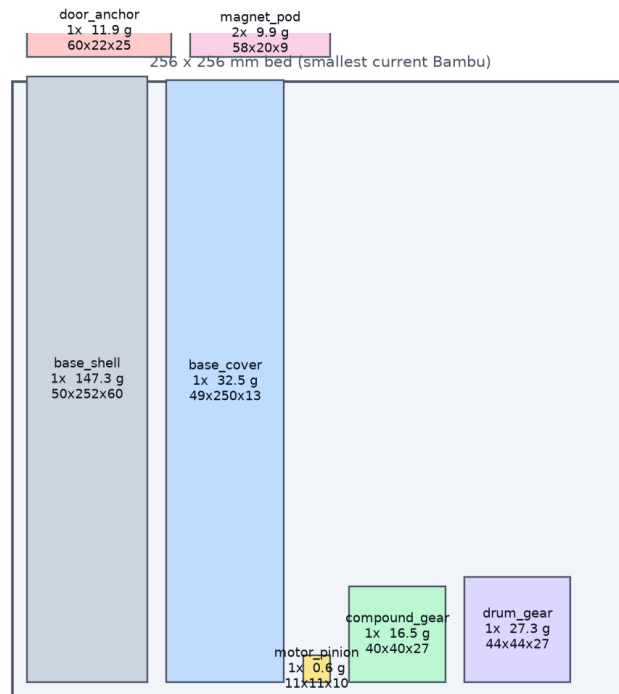
### 4 Something in the way

It refuses, retries every 19 s for a minute, then buzzes and says CLEAR THE WAY on the screen.

# Print these

7 designs, 8 pieces, 256 g. Every part lies flat, no supports.

## Everything you print



7 designs, 8 pieces, 256 g of PLA Tough+ at 100% infill. Every part prints flat with no supports; the shell is the only one that needs most of the bed.

| part          | qty | size mm   | g     |
|---------------|-----|-----------|-------|
| base_shell    | 1   | 50x252x60 | 147.3 |
| base_cover    | 1   | 49x250x13 | 32.5  |
| motor_pinion  | 1   | 11x11x10  | 0.6   |
| compound_gear | 1   | 40x40x27  | 16.5  |
| drum_gear     | 1   | 44x44x27  | 27.3  |
| door_anchor   | 1   | 60x22x25  | 11.9  |
| magnet_pod    | 2   | 58x20x9   | 9.9   |

Slice settings that matter: 0.2 mm layers, 4 walls, 40% infill everywhere except the gears and the nose, which want 100%. No supports needed on any part. Print the gears and the nose in PLA Tough+ for impact; if you have it, PLA Basic or PETG is actually stiffer for the anchor.

**ONE PAUSE:** magnet\_pod needs a print pause at the pocket roof. Drop both magnets in the same way up, then let it bridge three layers over them. No adhesive, and the magnets cannot creep out.

# Buy these

23 line items, 46 pieces, \$124.49 at the prices read on 2026-09-12

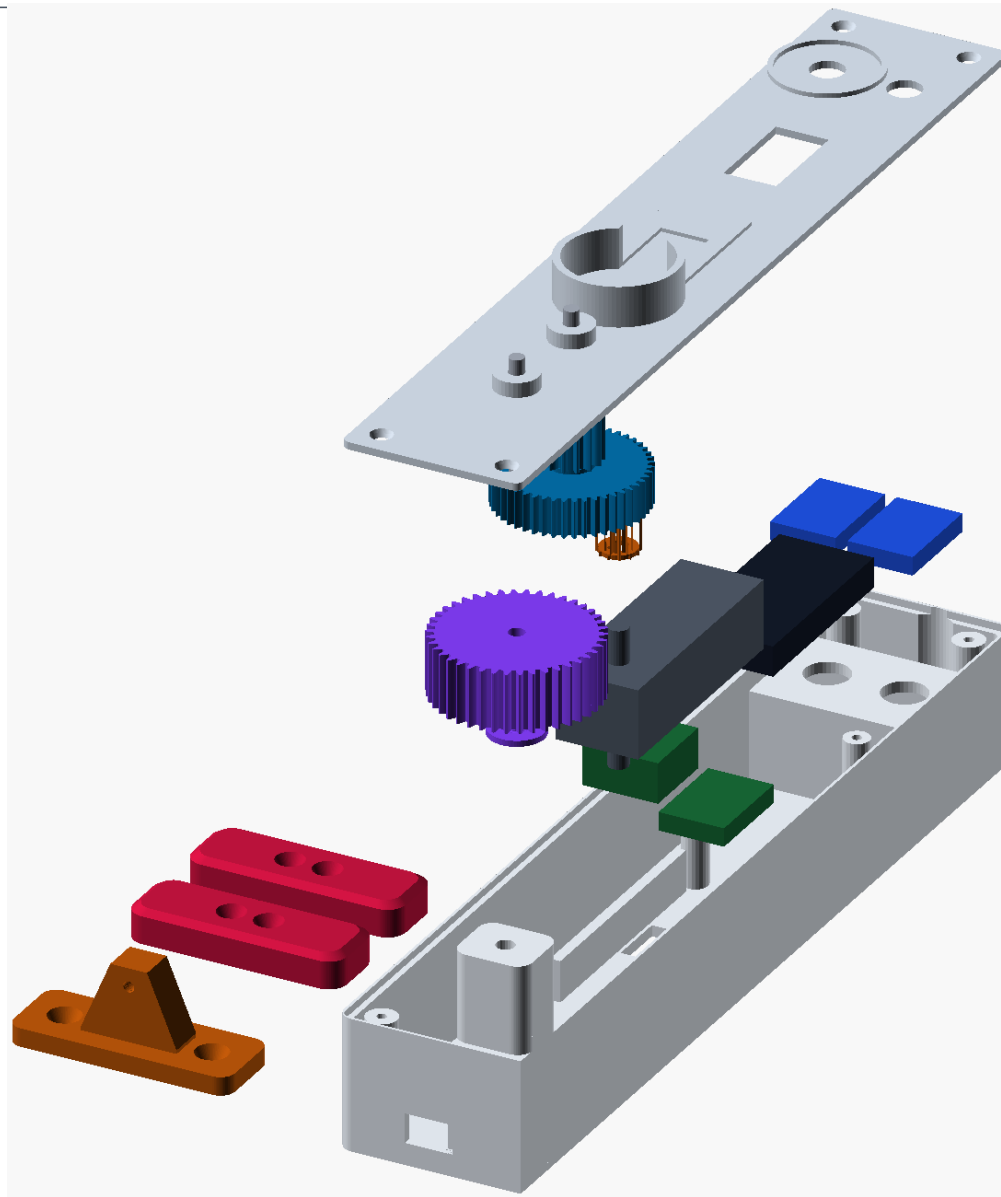
| vendor   | sku       | item                                       | qty | \$    | vendor        | sku            | item                                       | qty | \$    |
|----------|-----------|--------------------------------------------|-----|-------|---------------|----------------|--------------------------------------------|-----|-------|
| Adafruit | 3777      | DC gearbox 'TT' motor, 1:48, 3-6 V         | 1   | 2.50  | K&J Magnetics | D84            | Neodymium disc magnet, 12.7 x 6.35 mm, N   | 4   | 8.12  |
| Adafruit | 3297      | DRV8833 DC motor driver breakout           | 1   | 5.95  | generic       | dyneema-150    | Braided Dyneema line, 1.2 mm, 150 lb test, | 1   | 8.00  |
| Adafruit | 5396      | VL53L4CD time-of-flight distance sensor    | 1   | 14.95 | generic       | dowel-4x35     | Steel dowel pin, 4 x 35 mm                 | 2   | 1.20  |
| Adafruit | 3967      | VL53L1X time-of-flight distance sensor     | 1   | 14.95 | generic       | screw-8x32     | Wood screw, #8 x 32 mm, countersunk        | 3   | 0.45  |
| Adafruit | 2809      | LIS3DH triple-axis accelerometer           | 1   | 4.95  | generic       | screw-8x25     | Wood screw, #8 x 25 mm, countersunk        | 3   | 0.45  |
| LilyGO   | T-Display | TTGO T-Display ESP32, 1.14 in ST7789 135x2 | 1   | 15.00 | generic       | screw-6x20     | Wood screw, #6 x 20 mm, countersunk        | 4   | 0.48  |
| Adafruit | 3782      | TT motor encoder wheel, 20 slots           | 1   | 0.95  | generic       | m3x10          | M3 x 10 mm self-tapping screw              | 8   | 0.80  |
| Adafruit | 3986      | T-slot photo interrupter                   | 1   | 2.95  | generic       | m3x8           | M3 x 8 mm self-tapping screw               | 4   | 0.40  |
| Adafruit | 1536      | Piezo buzzer, 12 mm                        | 1   | 0.95  | Bambu Lab     | pla-tough-plus | PLA Tough+ filament, 1 kg spool            | 1   | 27.99 |
| Adafruit | 4399      | STEMMA QT / Qwiic JST SH 4-pin cable, 100  | 3   | 2.85  |               |                |                                            |     |       |
| Adafruit | 3564      | Silicone hook-up wire, 26 AWG, spool       | 1   | 4.95  |               |                |                                            |     |       |
| Adafruit | 4298      | USB-C cable, 1 m                           | 1   | 3.95  |               |                |                                            |     |       |
| Adafruit | 753       | 100 uF 16 V electrolytic capacitor         | 1   | 0.95  |               |                |                                            |     |       |
| Adafruit | 753-c     | 0.1 uF ceramic capacitor                   | 1   | 0.75  |               |                |                                            |     |       |

Two substitutions worth knowing. Adafruit 3875 is an ELECTROMAGNET, not a permanent magnet: it would need 0.6 A held continuously and the door would fall open on a power cut, so the design uses four K&J D84 permanent discs instead. Adafruit's own permanent disc (product 9) is out of stock and has no published pull force, which is why it is not specified here.

The motor's 5 V comes off the same USB source as the T-Display but is taken AT THE DRIVER. The board's only path from USB VBUS to its 5 V header pin is one unfused 1 A Schottky that also feeds the 3.3 V regulator, so a 1 A motor step on that node resets the ESP32.

# Exploded view

Colours key to the list on the right

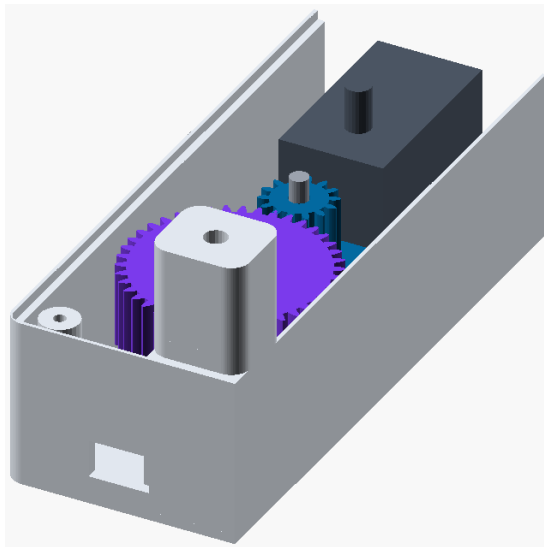


- |                                                                                     |                                               |                                |
|-------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------|
|  | <b>1. base_shell</b>                          | the unibody                    |
|  | <b>2. base_cover</b>                          | cover, windows and wave target |
|  | <b>3. TT motor (bought)</b>                   |                                |
|  | <b>4. motor_pinion</b>                        | 12T on the motor shaft         |
|  | <b>5. compound_gear</b>                       | 48T + 14T, one print           |
|  | <b>6. drum_gear</b>                           | 42T + cable drum, one print    |
|  | <b>7. 4 mm steel dowels (bought)</b>          |                                |
|  | <b>8. T-Display (bought)</b>                  |                                |
|  | <b>9. driver and accelerometer (bought)</b>   |                                |
|  | <b>10. two time-of-flight boards (bought)</b> |                                |
|  | <b>11. door_anchor</b>                        | on the door face               |
|  | <b>12. magnet_pod</b>                         | x2: door edge and jamb         |

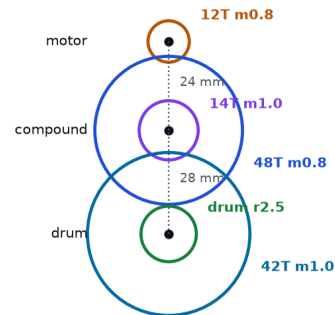
# Assemble the unit

About 30 minutes. Nothing is glued.

## Drive train: two printed spur stages, 12:1



cut away at the mid line



|                           |           |
|---------------------------|-----------|
| motor at the 1.0 A chop   | 48.2 mN·m |
| after stage 1 (48/12)     | 174 mN·m  |
| after stage 2 (42/14)     | 469 mN·m  |
| total ratio               | 12:1      |
| cable tension wound       | 151 N     |
| delivered past the chute  | 133 N     |
| cable travel, 90° to shut | 101 mm    |
| turns on the drum         | 5.20      |

### 1 Press the two dowels in

4 mm steel into the two floor bosses, square to the floor.

### 2 Pinion onto the motor

Push the 12T pinion onto the double-D shaft, flat to flat, until it bottoms.

### 3 Encoder wheel on the other shaft

It snaps on. The slot sensor goes in the pocket under the cover dome.

### 4 Motor into the cradle

Gearbox end down, leads towards the driver pocket. The cover rib holds it down.

### 5 Drum gear, then compound gear

Drum gear on the lower dowel, compound gear on the upper. Spin both by hand: no rub, no slop.

### 6 Boards into their pockets

Two M3 each. T-Display last, screen towards the cover window.

### 7 Wire it

Next page. Star the grounds at the driver.

### 8 Cable onto the drum

Thread through the radial hole, knot it in the flange pocket, wind on 5 turns, feed the tail out through the nose.

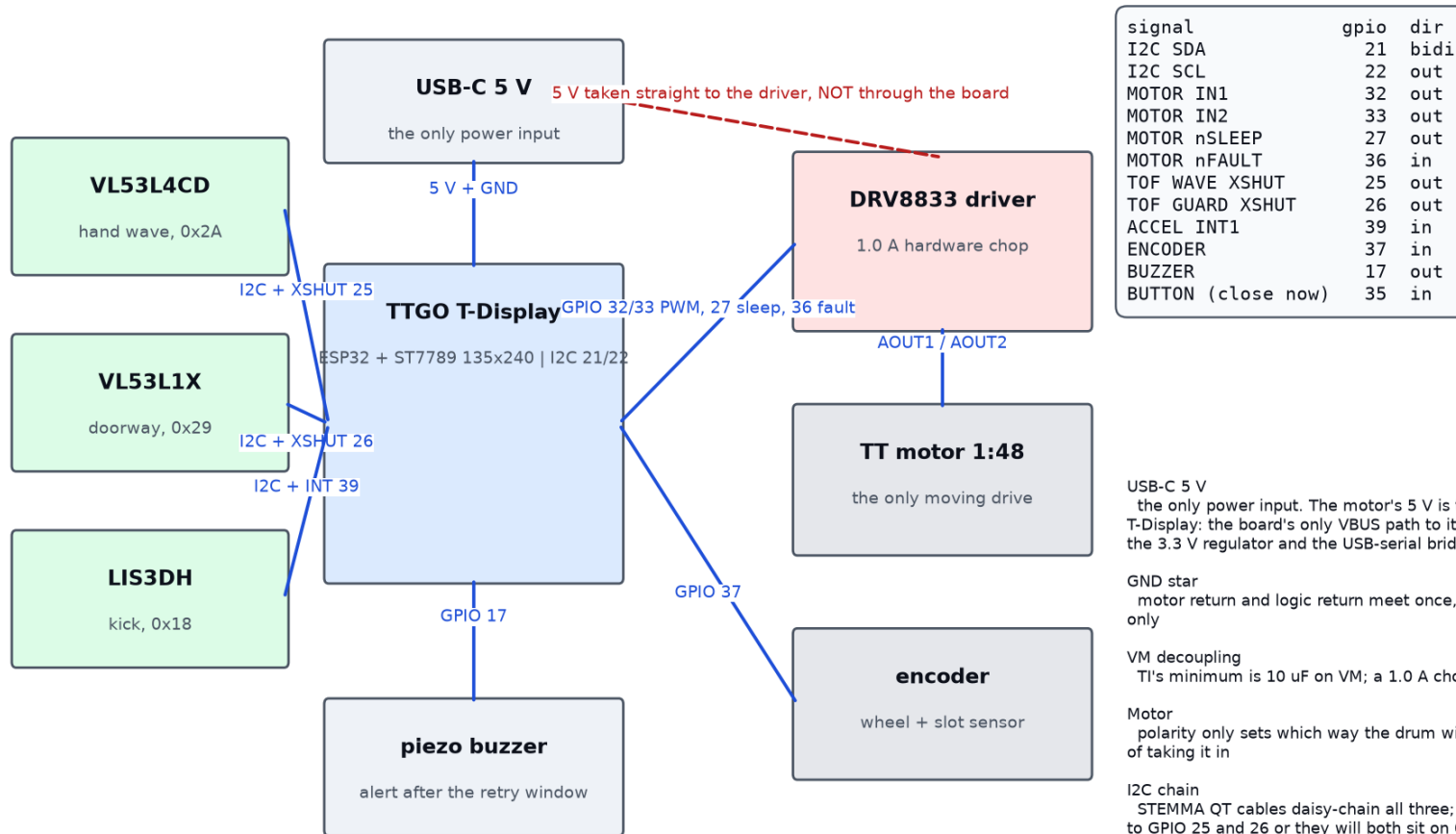
### 9 Cover on

Four M3. Check the dowel tops entered their blind bosses.

# Wire it

Twelve signals. Nothing touches a display pin, a flash pin or a strapping pin.

## Wiring: twelve signals, three I2C devices, one motor



**USB-C 5 V**  
the only power input. The motor's 5 V is taken from the same USB source AT THE DRIVER, not through the T-Display: the board's only VBUS path to its 5V header pin is a single 1 A 1N5819, unfused, and it also feeds the 3.3 V regulator and the USB-serial bridge, so a 1 A motor step on that node browns out the ESP32

**GND star**  
motor return and logic return meet once, at the driver. The T-Display header GND carries logic current only

**VM decoupling**  
TI's minimum is 10 uF on VM; a 1.0 A chopped stall on a shared USB rail earns more

**Motor**  
polarity only sets which way the drum winds. Swap the two leads if the first close pays cable out instead of taking it in

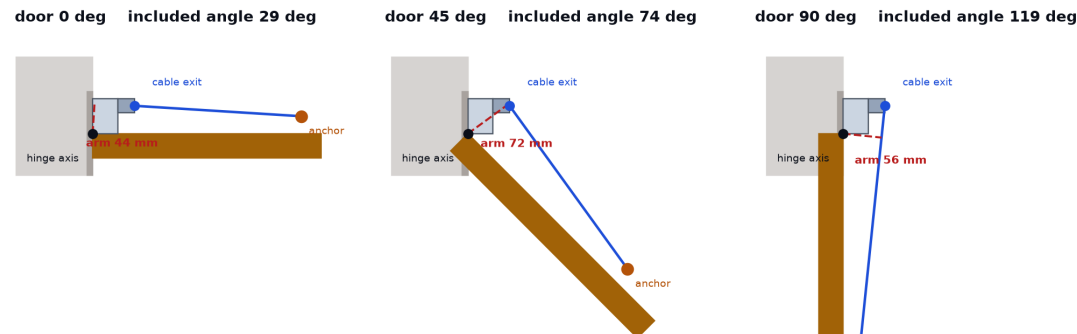
**I2C chain**  
STEMMA QT cables daisy-chain all three; the two VL53 boards also need their XSHUT lines run individually to GPIO 25 and 26 or they will both sit on 0x29 and fight

spare GPIO: [2, 12, 13, 15, 38]

# Install it

Hinge side, push side, behind the door. Three holes in the jamb, three in the door, two per magnet pod.

How it pulls: plan view at three door angles



The exit sits 40 mm off the closed-door plane on purpose. Put it ON that plane and at 0 deg the exit, the hinge axis and the anchor are in a straight line: the arm is exactly zero and no tension can shut the last degree. Offset, the included angle stays between 29 deg and 119 deg and the arm never drops below 44 mm.

## 1 Unit on the jamb reveal

Push side, hinge side, hand height. Use the jamb template. It stands 36 mm into the opening, inside the 51 mm rebate.

## 3 Magnet pods

One on the latch edge of the door, one on the jamb opposite. Faces towards each other, about 2 mm apart when shut.

## 5 Power

USB-C into the T-Display, motor 5 V taken at the driver.

## 2 Anchor on the door face

300 mm from the hinge axis, same height as the nose. Use the door template.

## 4 Cable

From the nose to the anchor eye. Taut with the door shut, no slack loop to catch on.

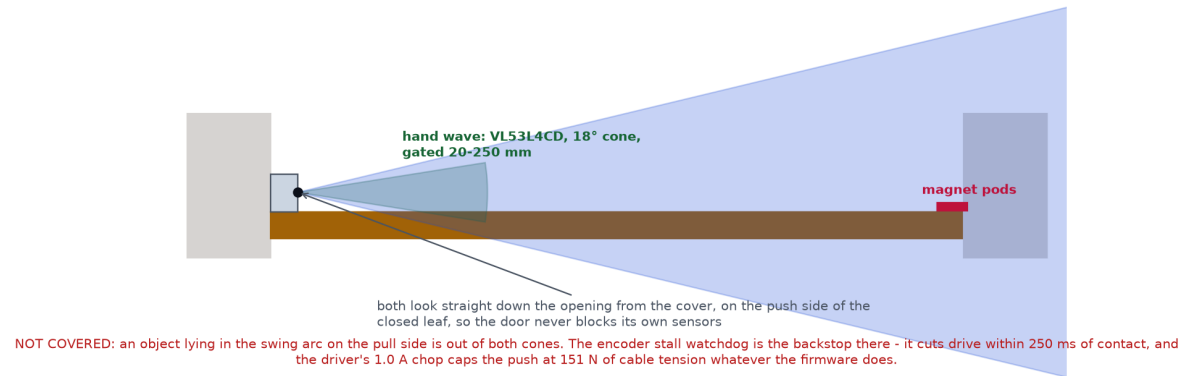
The cable and the whole unit sit on the push side of the closed leaf, so the door sweeps a quadrant that contains none of it. Nothing is ever in the door's path, and the door still opens fully by hand with the power off.

# First run

Flash it, then two things need setting on the actual door

## What the two time-of-flight sensors can see

doorway: VL53L1X, 27° cone,  
gated 120-1800 mm



### 1 Flash the firmware

pio run -e tdisplay -t upload from doorbot/firmware, over the T-Display's USB-C. It brings both time-of-flight boards up one at a time, moves the wave sensor to 0x2A, and refuses to drive the motor at all if either sensor is missing - the screen says which one.

### 2 Let it find the shut position

Press the top button with the door nearly shut. It winds until the leaf stops against the magnets, and that stall becomes its zero. The screen shows HOMING until then.

### 3 Set the kick threshold

Nobody publishes how much of a kick reaches a door jamb, so this one has to be measured on your door. Start at the shipped value, kick twice, and if nothing happens lower CLICK\_THS in setup(); if the door closes when someone shuts a cupboard, raise it.

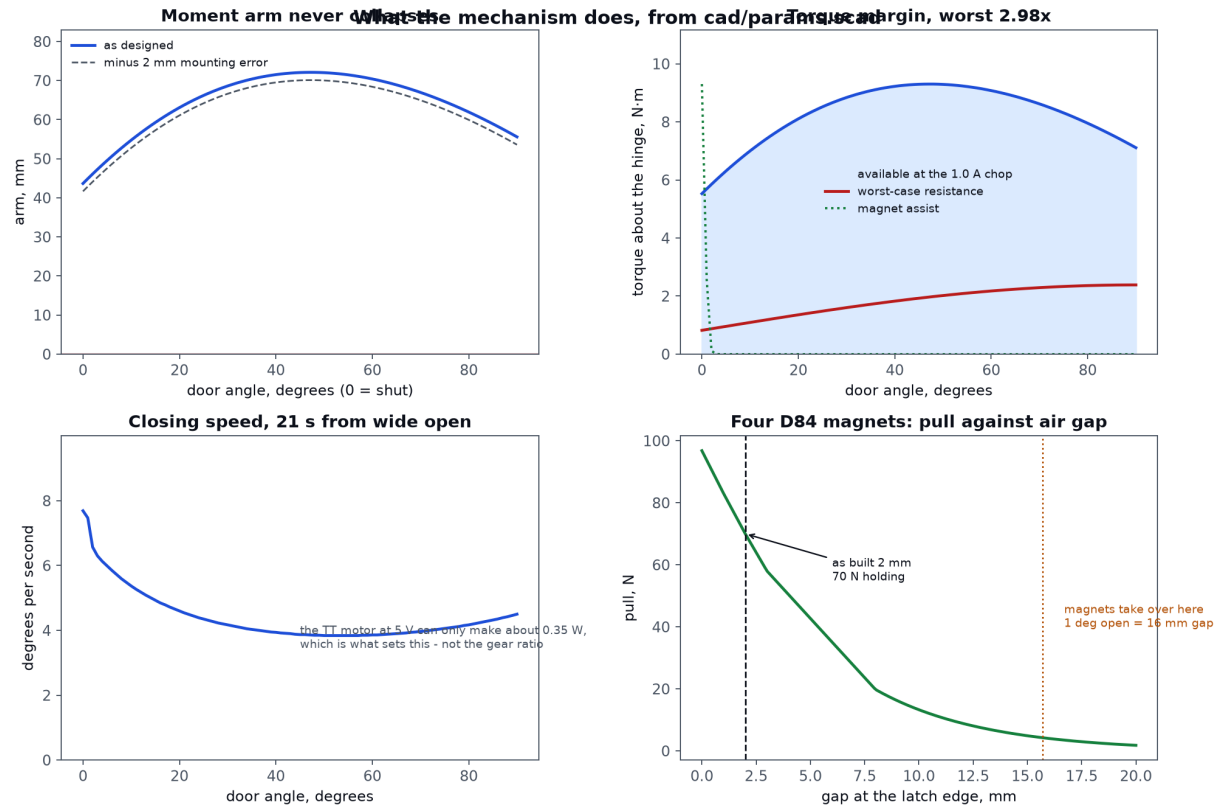
### 4 Check the wrong-direction case

If the first close pays cable out instead of taking it in, swap the two motor leads.

What the sensors cannot see: an object lying in the swing arc on the pull side is outside both cones. The encoder stall watchdog is the backstop - drive is cut within 250 ms of contact, and the driver's own 1.0 A chop caps the push at 151 N of cable tension whatever the firmware does.

# Why it works

Every number here comes from `cad/params.scad` through `scripts/check_mechanism.py`



The one thing that makes or breaks a hinge-side cable closer: if the cable leaves the jamb ON the closed-door plane, then with the door shut the exit, the hinge axis and the anchor are in a straight line. The moment arm is exactly zero and no tension at all can shut the last degree. Offsetting the exit 40 mm off that plane keeps the included angle between 29 and 119 degrees, and the arm between 42 and 54 mm.

The close takes 21 s, and no gear ratio fixes that: the TT motor at 5 V makes about 0.35 W of mechanical power, full stop. The upside is that the leaf arrives carrying 0.60 J against the 1.69 J that ANSI/BHMA A156.19 allows a closer.

24 mechanism checks pass, 18 CAD checks pass, and 7 verification steps pass in one command: `python scripts/verify.py`.

# Limits

What this package does not establish

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- **No prototype exists**

Everything in this guide is analysis of the design and a build of the firmware. Nothing here has been printed, assembled or run against a real door.

- **The hinge friction figure is modelled, not measured**

No published figure exists for plain residential butt hinges. The 0.82 N.m used throughout is the worst case of a friction model, and the torque margin quoted is against that model.

- **The kick threshold has no published basis**

No measurement of what a kick delivers to a door jamb exists in the literature searched. The shipped register value is a starting point to be calibrated.

- **Magnet figures are the vendor's**

The 70 N at a 2 mm gap comes from K&J's published pull-force curve for the D84, scaled by their own on-axis fall-off. It has not been measured on a door.

- **The pull-side swing arc is not sensed**

Two time-of-flight cones cover the doorway, not the arc on the pull side. Contact there is caught by the stall watchdog, not avoided.

- **Heavier or sprung doors are out of scope**

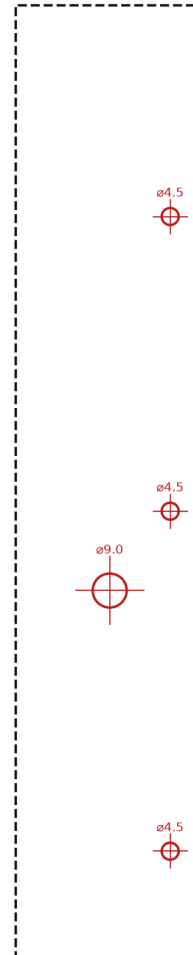
Sized for a 914 mm solid-core leaf at 40 kg on plain butt hinges with no closer spring. A sprung door or a fire door is a different machine.

# Template: jamb reveal

Print this page at 100%, then check the 50 mm bar with a ruler before you drill

---

## Jamb reveal: 3 holes



50 mm - check this is 50 mm before drilling

Hinge-side jamb reveal, push side. The 9 mm hole clears the motor's spare shaft; drill it 12 mm deep. Tape this page to the jamb with the bottom edge level.

## Template: door face

Print this page at 100%, then check the 50 mm bar with a ruler before you drill

---

### Door face: 3 holes



50 mm - check this is 50 mm before drilling

Anchor centre goes 300 mm from the hinge axis, at the same height as the unit's nose.

## Template: magnet pods

Print this page at 100%, then check the 50 mm bar with a ruler before you drill

---

### Magnet pod: 2 holes (x2, door edge and jamb)



50 mm - check this is 50 mm before drilling

One pod on the door's latch edge, one on the jamb directly opposite,  
magnet faces towards each other.